Sam McClenaghan

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Education

University of Victoria — Victoria, BC
Honours Bachelor of Science — *Sep 2023 – Present*
Computer Science major. Mathematics minor. GPA 7.8/9. Director of Projects at VikeLabs.

Experience

Software Engineer, Intern — May 2026 – Present
Vertical City — *Victoria, BC*

- Built an in-house video ad pipeline on AWS (S3, MediaConvert, CloudFront) serving millions of videos a day
- Reduced vendor bandwidth costs by 90% using VMAF-guided encoder tuning to compress each ad 5x
- Implemented perceptual hashing with ffmpeg and pHash to detect duplicate ads that survive re-encoding

Full-Stack Developer, Intern — May 2025 – Dec 2025
Leanpub — *Remote (Victoria, BC)*

- Shipped features end-to-end in a large Rails monolith spanning models, background jobs, and the front end
- Rebuilt the Dropbox integration on AWS S3 and background job queues to sync and publish author files
- Implemented Redis-backed background job progress tracking with real-time completion and failure states
- Hardened critical production flows with RSpec and FactoryBot test coverage of models and high-impact features

Software Developer, Intern — Nov 2022 – Aug 2024
Trust Science — *Edmonton, AB*

- Designed a version-control system for client scoring configurations using JSON Patch to store changes as deltas
- Built an Elasticsearch logging pipeline on DynamoDB streams to index and search logs from ML scoring jobs
- Migrated a 40+ page front end from Vue 2 and Bootstrap to Vue 3 and Tailwind CSS

Projects

CourseFlow | *Go, React, MySQL, Docker, DigitalOcean* | courseflow.smccl.ca — 2025 – 2026

- Built a course planner used by thousands of UVic students with live enrollment data and shareable timetables
- Designed a Go and MySQL backend that ingests live enrollment data and computes conflict-free timetables
- Developed the React scheduler with course search and interactive lecture, lab, and tutorial selection

Anchor | *Next.js, TypeScript, Supabase, Anthropic SDK, Resend* — 2025

- Built a mental-health companion at the UVicHacks × Claude hackathon that detects distress from HRV data
- Integrated the Anthropic SDK to deliver Claude-powered grounding support during detected distress episodes
- Owned backend and DevOps: Supabase/Postgres migrations, Vercel deployment, and Garmin data ingestion

Technical Skills

Languages: Ruby, Go, TypeScript, JavaScript, Python, Java, C, SQL
Frameworks: Ruby on Rails, RSpec, React, Next.js, Remix, Vue, Tailwind CSS
Cloud & Tools: AWS (S3, Lambda, MediaConvert, CloudFront, DynamoDB), Docker, ffmpeg
Databases: MySQL, PostgreSQL, Redis, Elasticsearch